

Draft: thermographs, Newton polygons, and the missing residue

a9lim

July 2026

Abstract

This note pursues `docs/OPEN.md` problem `under*(e_g~e_s)`: whether the two tropical consumers in `ogdoad`—game thermography and valuation/Newton polygons—are one tropical object or merely a convention mirror. The result of the first pass was a negative theorem at the thermograph level and a more precise positive target: a temperature associated graded with residues.

The negative theorem is small but decisive: the thermograph is not a congruence for disjunctive sum. There is no binary operation, single- or multi-step, that takes only $\text{Th}(G)$ and $\text{Th}(H)$ and returns $\text{Th}(G + H)$ for all short games. The witness is already in the temperature-zero layer: $\text{Th}(*) = \text{Th}(\uparrow)$, but $* + * = 0$ while $\uparrow + *$ is still a non-number of temperature 0.

The useful replacement is not “no tropical structure”. Temperature itself has exactly the expected tropical-valuation shape: $\text{temp}(G + H) \leq \max(\text{temp } G, \text{temp } H)$, with equal-temperature pairs forming the game-side vanishing locus. What fails is the proposed angular component $(\text{mean}(G), \text{temp}(G))$: it forgets the leading thermic residue. A single object, if it exists, should be an associated graded object of the game group by temperature, enriched at least by the all-small residue data (atomic weight plus its kernel), not the ordinary thermograph alone.

The second pass, after `src/games/heating.rs` shipped game-valued heating, Berlekamp overheating, and Norton multiplication, tests the multiplicative hope directly. The unrestricted answer is negative. In the $\tau = 0$ quotient, $*$ and $* + 1$ differ by a cold number, but Norton multiplication by the positive infinitesimal unit $\uparrow$ converts that hidden integer into a leading temperature-zero residue. The same obstruction appears for the degenerate overheating operator $\int_{\uparrow}^0$. Thus the full Berlekamp/Norton operator does not descend to the naive associated graded.

The third pass settles the leading restricted case positively. If the Norton unit is the positive dyadic number $u = m/2^k$, put $\delta = 2^{-k}$ (and $\delta = 1$ for an integer) and $a = u - \delta$. Then every nonnumeric thermographic game obeys

$$\text{mean}(G.u) = u \text{mean}(G), \quad \text{temp}(G.u) = u \text{temp}(G) + a.$$

A cold dyadic number can become hot under Norton multiplication, but its exact temperature is always strictly below a . Norton linearity therefore induces an additive map $\text{gr}_{\tau} \rightarrow \text{gr}_{u\tau+a}$ for every $\tau \geq 0$; equivalently the matching Berlekamp operator $\int_u^a$ descends. For integer $u = n$, the shifted thermal height $h = \tau + 1$ simply scales as $h \mapsto nh$. Thus the mirror is not purely notational: the game filtration has a genuine family of numeric regrading transports, although no multiplicative action, unrestricted internal product, or complete residue model is claimed.

Claim levels. **[PROVED]** – proof from standard short-game identities plus the live implementation’s definitions; **[TESTED]** – verified by the commands in Appendix A; **[STANDARD MATH]** – standard thermography/Newton-polygon math used by the repo; **[CONJECTURAL]** – plausible but not proved here; **[OPEN]** – this pass did not solve it.

1 The two tropical objects currently in the code

[**STANDARD MATH**] The place axis is the usual non-Archimedean tropicalization. In `src/scalar/valued.rs`, a valued scalar satisfies

$$v(xy) = v(x) + v(y), \quad v(x + y) \geq \min(v(x), v(y)),$$

with equality when $v(x) \neq v(y)$. The adaptor `tropicalize` lands in the min-plus semiring. In `src/scalar/newton.rs`, $\text{NP}(f)$ is the lower convex hull of the points $(i, v(a_i))$, and its side slopes are the negatives of root valuations. The Springer tests in `src/forms/springer/local.rs` check that the Newton polygon forgets only the residue square-class data: its slopes are exactly the valuation layers.

[**STANDARD MATH**] The game axis is also tropical, but internally. A short game's thermograph consists of a left wall $L_G(t)$, a right wall $R_G(t)$, the mast $\text{mean}(G)$, and the temperature $\text{temp}(G)$; the implementation lives in `src/games/thermography.rs`. The tropical naming layer makes the left wall the max-plus fold over Left options' right walls, the right wall the min-plus fold over Right options' left walls, and cooling ordinary addition of the temperature parameter, i.e. tropical multiplication. The test `via_tropical_matches_thermograph` pins this naming equal to the ordinary thermograph.

Remark 1. This is already enough to justify the slogan “thermography is tropical arithmetic”. It is not enough to make thermographs compose under disjunctive sum. Newton polygons have Dumas additivity under polynomial multiplication. Thermographs do not have an analogous strict additivity under game sum.

2 The thermograph is not a sum invariant

Definition 1. For a short game G , write

$$\text{Th}(G) = (L_G, R_G, \text{mean}(G), \text{temp}(G))$$

for the exact thermograph computed by `thermograph`. A thermograph-level sum law would be any operation Φ such that

$$\text{Th}(G + H) = \Phi(\text{Th}(G), \text{Th}(H))$$

for all games in the domain of ordinary thermography.

Proposition 1 (No thermograph-level composition law). [**PROVED**] *The map $G \mapsto \text{Th}(G)$ is not a congruence for disjunctive sum. Consequently, no operation whose inputs are only $\text{Th}(G)$ and $\text{Th}(H)$ can determine $\text{Th}(G + H)$ for all short games.*

Proof. Use the standard games $* = \{0 \mid 0\}$ and $\uparrow = \{0 \mid *\}$. The live thermography engine, and standard temperature theory, give

$$\text{Th}(*) = \text{Th}(\uparrow) = (0, 0, \text{mean} = 0, \text{temp} = 0),$$

where both walls are the constant zero wall.

Now add the same partner $*$. Since $*$ is the number of order two,

$$* + * = 0,$$

so the result is a number and the repo reports temperature -1 for the cold numeric case. But $\uparrow + *$ is still all-small and non-numeric; its thermograph has temperature 0 and constant zero walls. Thus the two input pairs

$$(\text{Th}(*), \text{Th}(*)) \quad \text{and} \quad (\text{Th}(\uparrow), \text{Th}(*))$$

are identical, while the output thermographs differ. This contradicts the existence of Φ . $\square$

Corollary 1. [PROVED] *The pair $(\text{mean}(G), \text{temp}(G))$ cannot be the game-side analogue of $(\text{ac}(x), v(x))$ for a valued field element. It forgets the leading thermic residue that decides cancellation.*

Remark 2. The witness is deliberately low-tech. It lives at temperature zero, before any subtle sidling or nested hot-game pathology appears. The obstruction is not an implementation artifact of complicated thermographs; it is present in the first infinitesimal layer.

3 What survives: temperature as a tropical valuation

[STANDARD MATH] Temperature does satisfy the valuation-shaped inequality

$$\text{temp}(G + H) \leq \max(\text{temp}(G), \text{temp}(H)),$$

with numbers treated as colder than all infinitesimal/hot games. Reversing the order by $v_T(G) = -\text{temp}(G)$ gives the familiar valuation form

$$v_T(G + H) \geq \min(v_T(G), v_T(H)).$$

The equal-temperature case is the game-side vanishing locus. It can cancel downward, or it can retain the same leading temperature.

[TESTED] In this checkout I probed a small catalogue consisting of integers, $*$, $\uparrow$, $\downarrow$, $*2$, switches $\{a \mid b\}$ for small integer a, b , and one-option nested hot games. Across 324 pairwise sums there were no violations of $\text{temp}(G + H) \leq \max(\text{temp } G, \text{temp } H)$. Every unequal-temperature pair hit equality. Equal-temperature pairs produced the expected downward spread:

$$\begin{aligned} \text{temp}(*) &= \text{temp}(*) = 0, & \text{temp}(* + *) &= -1, \\ \text{temp}(*) &= \text{temp}(\uparrow) = 0, & \text{temp}(* + \uparrow) &= 0, \\ \text{temp}(\{1 \mid -1\}) &= \text{temp}(\{2 \mid 0\}) = 1, & \text{temp}(\{1 \mid -1\} + \{2 \mid 0\}) &= -1. \end{aligned}$$

Other equal-temperature probes at input temperature 2 produced output temperatures $-1, 0, 1$, and 2 .

Remark 3. This is exactly the tropical-hyperfield shape if one keeps only the scalar temperature: unequal leading values add strictly by max-plus; equal leading values may cancel to any lower layer. The problem is that this scalar law is too coarse to recover walls, masts below the leading layer, or the resulting game value.

4 The one-object probe: switches versus one-side Newton polygons

[TESTED] There is a clean but shallow dictionary for switches. For $m \in \mathbb{Q}$ and $\tau > 0$, the switch

$$S_{m,\tau} = \{m + \tau \mid m - \tau\}$$

has

$$L_S(t) = m + \tau - t, \quad R_S(t) = m - \tau + t \quad (0 \leq t \leq \tau),$$

and both walls freeze to m at $t = \tau$.

The Newton polygon of a binomial $x^n - \pi^k$ has one side from $(0, k)$ to $(n, 0)$, hence one slope and one root-valuation layer. After an affine change of axes, one wall of $S_{m,\tau}$ is exactly such a one-side polygon; the other wall is its dual reflection. This realizes the requested “one-object probe” only in the trivial one-parameter family.

Proposition 2. [PROVED] *The switch/binomial dictionary does not extend to a thermograph-level sum theorem.*

Proof. The dictionary sends one switch to one linear Newton side. Polynomial multiplication concatenates Newton slope data by Dumas additivity. But Proposition 1 shows that disjunctive sum cannot be read from thermographs alone, even before leaving the zero-temperature layer. Any extension would need extra residue data not present in the switch wall. $\square$

5 The actual candidate bridge: an associated graded game object

[CONJECTURAL] The positive object should be a temperature filtration of the game group. Let

$$F_{\leq \tau} = \{G : \text{temp}(G) \leq \tau\}$$

for $\tau \geq 0$, with the numeric games placed in the cold bottom layer. The standard temperature inequality says these are additive subgroups. The associated graded candidate is

$$\text{gr}_T(\text{Games}) = \bigoplus_{\tau} F_{\leq \tau} / F_{< \tau}.$$

The leading thermic residue of G would be the class of G in the quotient at $\tau = \text{temp}(G)$, not the pair $(\text{mean}(G), \text{temp}(G))$.

[TESTED] At $\tau = 0$, this is already visible. The thermograph collapses $*$, $\uparrow$, $\downarrow$, and $*2$ to the same constant-zero object. Atomic weight, implemented in `src/games/atomic_weight.rs`, recovers one additive residue on all-small games:

$$aw(\uparrow) = 1, \quad aw(\downarrow) = -1, \quad aw(*) = 0.$$

But atomic weight still has a kernel containing nimber-like residues such as $*$, and $* + * = 0$ shows that this kernel matters. So even the first graded piece is not “the mast”. It is a genuine residual game object.

Remark 4. This is the closest analogue to the valued-field picture. The Newton polygon records valuations and forgets residues; the initial forms in $\text{gr}_v(K[x])$ restore them. The thermograph records temperatures and walls but forgets thermic residues; a graded game object would restore them. The difference is that $\text{gr}_v(K)$ is a graded ring, while short games under disjunctive sum are only an abelian group.

6 The Norton/overheating descent test

[**TESTED**] After the first pass, the repo gained `src/games/heating.rs`: game-valued heating, Norton multiplication $G.U$ by a positive unit U , and Berlekamp overheating $\int_s^t G$. This removes the old infrastructure excuse. We can ask directly whether these operators are well-defined on the associated-graded quotient.

For a fixed layer $\tau \geq 0$, write

$$G \equiv_\tau H \iff \text{temp}(G - H) < \tau.$$

At $\tau = 0$, the lower subgroup $F_{<0}$ is the cold numeric subgroup. Thus $*$ and $* + 1$ have the same residue in $F_{\leq 0}/F_{<0}$, since

$$* - (* + 1) = -1, \quad \text{temp}(-1) = -1 < 0.$$

Proposition 3 (Unrestricted Norton descent fails). [**TESTED**] *Norton multiplication by a positive nonnumeric unit does not descend to the naive temperature associated graded.*

Proof. Let $U = \uparrow$, which is strictly positive and has temperature 0. The live implementation reports

$$\text{temp}(*.U) = 0, \quad \text{temp}((* + 1).U) = 0,$$

but

$$\text{temp}((*U) - ((* + 1).U)) = 0, \quad \text{aw}((*U) - ((* + 1).U)) = -1.$$

The output difference is therefore not in the lower subgroup $F_{<0}$; it is a leading temperature-zero residue. So two equivalent representatives in the $\tau = 0$ quotient produce inequivalent outputs.

The mechanism is exactly the danger one would expect from the definition: Norton multiplication uses U on integer leaves. The hidden cold integer $+1$, invisible before passing to the quotient, becomes the visible temperature-zero game $\uparrow$ after applying the operator. $\square$

[**TESTED**] The same obstruction appears for overheating. With lower unit $s = \uparrow$ and upper shift $t = 0$,

$$\text{temp} \left(\int_{\uparrow}^0 * - \int_{\uparrow}^0 (* + 1) \right) = 0, \quad \text{aw} \left(\int_{\uparrow}^0 * - \int_{\uparrow}^0 (* + 1) \right) = -2.$$

Thus $\int_s^t$ also fails to descend when the lower unit is allowed to be a nonnumeric positive game.

[**TESTED**] The bounded probe `experiments/under_descent.py` records this explicit witness and also runs a small sanity scan for ordinary numeric units. On a 21-game catalogue it checked 126 representative pairs for Norton multiplication by units 1 and 2, and the same 126 pairs for $\int_s^0$ with $s = 1, 2$. It found no numeric-unit failures. This is not a theorem; it is a useful pointer that the obstruction is not “Norton exists, so everything fails”, but specifically that nonnumeric units can see cold coefficients erased by the naive quotient.

7 Positive result: numeric Norton units descend

The bounded scan above left one conspicuous possibility alive. It is not an accident of the small catalogue; numeric units admit an exact formula.

Definition 2. Let $U = u > 0$ be a short-game number, written in lowest dyadic form $u = m/2^k$. Put

$$\delta_U = \begin{cases} 2^{-k}, & k > 0, \\ 1, & k = 0, \end{cases} \quad a_U = u - \delta_U.$$

Thus the canonical options of a nonintegral u are $u - \delta_U$ and $u + \delta_U$, while a positive integer has the sole Left option $u - 1$.

Proposition 4 (exact numeric Norton thermics). **[PROVED]** *Let $U = u > 0$ be dyadic and write $a = a_U$. For every short game G in the finite thermographic domain,*

$$\text{mean}(G.u) = u \text{mean}(G).$$

If G is not a number, then

$$\text{temp}(G.u) = u \text{temp}(G) + a. \quad (1)$$

If $G = x$ is a number and $\epsilon_x = 1/\text{den}(x)$ is its canonical dyadic mesh, then

$$\text{temp}(x.u) = \begin{cases} -1, & a - u\epsilon_x < 0, \\ a - u\epsilon_x, & a - u\epsilon_x \geq 0. \end{cases} \quad (2)$$

In particular, the image of every cold number has temperature strictly less than a .

Proof. For the canonical numeric unit u , every Norton increment $U + I$ is the same number a : a Left option contributes $U^L = u - \delta_U = a$, and a Right option contributes $2U - U^R = u - \delta_U = a$. Hence the defining recursion is

$$n.u = nu \quad (n \in \mathbb{Z}), \quad G.u = \{G^L.u + a \mid G^R.u - a\} \quad (G \notin \mathbb{Z}). \quad (3)$$

We first need the numeric-option lemma hidden by the integer base case in (3). If x has exact denominator 2^j , induction on j gives $\text{mast } ux$. Integers are literal leaves and have temperature -1 . If $j > 0$, the canonical options are $x \pm 2^{-j}$ and have smaller denominator. Once their images have frozen, the parent walls meet at $s_x = a - u2^{-j}$. When $a = 0$ (precisely $u = 1$ or $u = 2^{-k}$), the simplicity theorem instead makes every $x.u$ the number ux ; when $a > 0$, one has $s_x \geq 0$ for every $j > 0$. Thus (2) follows, together with the strict bound $\text{temp}(x.u) < a$. In particular, Norton multiplication treats integer leaves specially, not all number leaves: for example $(1/2).2 = 1 + *$ has temperature 0.

Now let G be nonnumeric and induct on its birthday. At a scaffold parameter $s = a + ut$, $t \geq 0$, (3), translation of walls, and the induction hypothesis give

$$L_{G.u}(a + ut) = uL_G(t), \quad R_{G.u}(a + ut) = uR_G(t). \quad (4)$$

Numeric options cause no exception: by (2) they have already frozen strictly below a , with mast equal to their ordinary product by u . Equation (4) locates a meeting at $a + u \text{temp}(G)$; it remains to rule out a premature meeting below a .

Write $D_G(t) = L_G^{\text{raw}}(t) - R_G^{\text{raw}}(t) - 2t$. The wall-slope bounds make D_G nonincreasing. If $\text{temp}(G) > 0$, then $D_{G.u}(a) = uD_G(0) > 0$, so monotonicity rules out every earlier zero. Only the infinitesimal case $\text{temp}(G) = 0$ needs care. We use the following elementary stop lemma.

No-premature-meeting lemma. Let $a > 0$, let m be a number, and let X be canonical. If the transformed right wall of $X.u$ has maximal slope $+1$ up to height a and terminates at um , then

Right wins $X - m$ moving first. The mirror statement holds for a transformed left wall of slope -1 .

For completeness, prove the lemma by induction on the birthday of X . The maximal slope forces an extremizing Right option Y . If Y is numeric, the numeric-option lemma and its strict freeze below a force $Y = m$, giving the winning move immediately. Otherwise every Left reply Z of Y has $\text{RS}(Z) \leq m$. Strict inequality is already a Right win; equality forces the same maximal-slope condition and invokes the induction hypothesis. Number avoidance ensures that these are all relevant replies in $Y - m$. This proves the lemma; negation proves its mirror.

Suppose now that both transformed walls met on a terminal interval below a . Put $m = \text{LS}(G) = \text{RS}(G)$. Every stop-maximizing Left option then satisfies the first half of the lemma, while lower-stop options are immediate Right wins. Number avoidance therefore gives $G \leq m$; the mirror argument gives $G \geq m$. Hence $G = m$, contrary to G being nonnumeric. There is no premature meeting. Equation (4) therefore gives exactly $\text{temp}(G.u) = a + u \text{temp}(G)$ and common mast $u \text{mean}(G)$. $\square$

Corollary 2 (numeric Norton descent). **[PROVED]** *For every $\tau \geq 0$, Norton multiplication by a fixed positive dyadic unit u induces an additive map*

$$\overline{A}_{u,\tau} : F_{\leq \tau} / F_{< \tau} \longrightarrow F_{\leq u\tau + a_U} / F_{< u\tau + a_U}. \quad (5)$$

The same map is the generalized overheating operator $\int_u^{a_U}$. For a positive integer n , the shifted degree $h = \tau + 1$ satisfies

$$h(G.n) = n h(G).$$

Proof. Norton multiplication by a fixed positive unit is additive, a standard Norton product theorem. If $G - H$ is nonnumeric of temperature $\sigma < \tau$, (1) gives

$$\text{temp}((G - H).u) = u\sigma + a_U < u\tau + a_U.$$

If $G - H$ is a number, (2) instead puts its image strictly below a_U , hence strictly below the target layer. Thus the map is independent of the chosen representative and is additive. Finally, (3) is exactly Berlekamp overheating from lower unit u to upper shift a_U . For $u = n$, $a_U = n - 1$, so $\text{temp}(G.n) + 1 = n(\text{temp}(G) + 1)$ on every nonnumeric graded layer. $\square$

[TESTED] Proposition 4 is computed without materializing the Norton product by the two Rust helpers

`numeric_norton_regrade` `numeric_norton_mean_temperature`.

Source-level tests compare them with the actual recursive product on integer and nonintegral numeric games, all-small games, switches, and a nested hot game for eight units from $1/4$ through 3. The Python probe independently checked 210 game/unit thermic pairs and 24 representative pairs for both Norton multiplication and the matching overheating operator, with no numeric-unit failure.

Remark 5. This is the first nontrivial positive transport in the **under** thread. It is weaker than a ring structure: u is an external numeric coefficient, not an arbitrary game, and (5) regrades rather than multiplying two residual game classes. It is not even a multiplicative action of the positive dyadics: $A_2 A_{1/2}(\ast)$ has temperature 1, whereas $A_1(\ast) = \ast$ has temperature 0. The result is a family of additive transports. That is still stronger than the scalar temperature hyperoperation: each transport acts on the residual classes themselves and, for integer units, multiplies thermal height exactly as a ramification index multiplies a valuation scale.

8 What remains after the third pass

[OPEN] I do not currently have an *internal* product on $\text{gr}_T(\text{Games})$ that would make it the literal peer of the graded ring of a valued field. The old candidate “use Norton multiplication / overheating” is now too broad: Proposition 3 rules out unrestricted nonnumeric units on the naive quotient. Corollary 2 proves that every positive numeric unit survives, so the boundary is now exact rather than experimental. Full short games still do not form a ring, and the repo correctly refuses to pretend otherwise.

[OPEN] I also do not have a nontrivial thermograph hyperoperation. The scalar temperature hyperoperation is clear and useful:

$$a \boxplus b = \begin{cases} \max(a, b), & a \neq b, \\ \{\text{temperatures} \leq a\}, & a = b, \end{cases}$$

with the usual convention that colder numbers lie below 0. But lifting this from scalar temperatures to full thermographs without adding residues becomes either false (Proposition 1) or tautological (return “all thermographs of all sums of representatives”).

[OPEN] The next formulation is therefore not “find a binary operation on thermographs”, and not “use the full Berlekamp/Norton surface”. Numeric units are solved. What remains is:

Define a residue-enriched temperature quotient for a manageable class of games and decide whether it supports anything beyond the proved external numeric regrading: mean-normalized nonnumeric units with cold coefficients retained separately, or an all-small quotient refined by atomic weight plus its nimber-like kernel.

The answer is already no longer “purely notational”: (5) is a substantive filtered-group transport. The stronger “one tropical ring” reading remains open.

A Checks run in this pass

The code-level facts used above were checked against the live repo with:

- `cargo test tropical -- --nocapture`: 15 relevant tests passed, including tropical semiring laws and `thermograph_via_tropical`.
- `cargo test newton -- --nocapture`: 8 Newton-polygon tests passed, including Eisenstein slopes and Dumas additivity.
- `cargo test springer -- --nocapture`: 50 Springer/local-global tests passed, including the Newton-polygon/Springer-shadow check.
- A Python probe through the installed `.venv` binding checked 324 small-game sums for the temperature inequality and the off-diagonal equality pattern, and printed the witnesses in Proposition 1.
- `cargo test heating`: 15 heating/Norton/overheating tests passed, including the explicit non-descent witness `hot_units_do_not_descend_mod_cold_numbers`, the exact dyadic regrade, the load-bearing numeric-image lemma, matching-overheating equality, numeric quotient descent, a bounded day-three singleton-option census, a Norton-linearity sentinel, and the failure of multiplicative composition.
- `cargo check --features python`: the new Python thermic shortcuts compile against the feature-gated bindings.

- `python experiments/under_descent.py`: recorded the $*$ versus $* + 1$ obstruction for unit $\uparrow$; checked 210 exact thermic predictions on a 35-game catalogue and six numeric units; and checked 24 representative pairs each for Norton multiplication and matching overheating, with zero numeric-unit failures.

References

- [1] Berlekamp, Conway, and Guy. *Winning Ways for your Mathematical Plays*.
- [2] Aaron Siegel. *Combinatorial Game Theory*.
- [3] Elwyn Berlekamp. The economist’s view of combinatorial games. In *Games of No Chance*, 1996.
- [4] Elwyn Berlekamp. Blockbusting and Domineering. *Journal of Combinatorial Theory, Series A* 49 (1988), 67–116.
- [5] Oleg Viro. Hyperfields for tropical geometry I.
- [6] Diane Maclagan and Bernd Sturmfels. *Introduction to Tropical Geometry*.